

House Price Prediction Using Machine Learning and Explainable AI

Executive Summary

This project focused on developing a machine learning framework to predict residential property prices using housing characteristics from the Housing Prices Dataset. The primary objective was to estimate house prices accurately while identifying the factors that have the greatest influence on property valuation. The dataset contained information related to area, number of bedrooms, bathrooms, stories, parking availability, furnishing status, air conditioning, basement facilities, preferred location status, and other housing attributes.

The project began with data exploration and preprocessing. Missing values and duplicate records were identified and handled appropriately to improve data quality. Categorical variables were transformed into numerical representations using one-hot encoding, making the dataset suitable for machine learning algorithms. Additional feature engineering techniques were applied by creating new variables such as `total_rooms` and `area_per_room` to capture hidden relationships between housing characteristics and property prices.

A comprehensive model comparison study was conducted using seven regression algorithms: Linear Regression, Random Forest Regressor, Gradient Boosting Regressor, Decision Tree Regressor, K-Nearest Neighbors Regressor, Support Vector Regressor, and XGBoost Regressor. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. Among all evaluated models, Linear Regression achieved the highest predictive performance with an R^2 Score of 0.659, followed closely by XGBoost (0.646) and Gradient Boosting Regressor (0.618). These results indicate that the relationship between housing features and property prices within the dataset is largely linear, making Linear Regression the most suitable model for this problem.

Several visualizations were created to better understand the data and model behavior, including house price distribution plots, correlation heatmaps, actual versus predicted price comparisons, feature importance rankings, residual analysis, and model comparison charts. To improve model transparency and interpretability, SHAP (SHapley Additive Explanations) was implemented. The SHAP analysis identified `area` as the most influential feature affecting house prices, followed by `total_rooms`, `area_per_room`, air conditioning availability, number of bathrooms, preferred area status, furnishing status, basement availability, and parking capacity.

The findings demonstrate that larger properties with more rooms, better amenities, and favorable locations tend to command significantly higher prices. The project highlights the effectiveness of machine learning in supporting real estate valuation and decision-making processes. From a business perspective, real estate organizations can leverage these insights to improve pricing strategies, prioritize high-value property features, and make more informed investment decisions.

Overall, the project successfully combined predictive modeling, feature engineering, model comparison, explainable artificial intelligence, and business-oriented analysis to create a robust and interpretable house price prediction system.